AI

TECHNICAL WALKTHROUGH

AI Use Case Portfolio Analyzer

The data contract, the gates, the scoring, and the shared-layer algorithm.

Python 3.9+ · no dependencies · local model optional

The ontology names six classes and seven properties.

Class	Meaning
AIUseCase	The unit being governed.
BusinessOwner	The accountable person.
DataSource	An approved system the use case may read.
Entity	A business object such as customer or document.
RiskTier	Low, moderate, high, critical.
Control	Human-in-the-loop, monitoring.
Bucket	Now, Next, Later, Hold.

Properties connect them: hasBusinessOwner, readsFrom, requiresEntity, providesEntity, hasRiskTier, hasControl, assignedToBucket.

ONTOLOGY

The ontology is not documentation. It is the control plane.

- Risk tiers live in the ontology, so the gate reads them rather than hard-coding them.
- The governance gates reference properties: named owner, human-in-the-loop, monitoring, approved source.
- Approved data sources are data, not code, so the policy can change without a release.
- Data sources map to entities, which is what makes the shared-layer analysis possible.
- Change the ontology and the behaviour changes. The analyser stays the same.

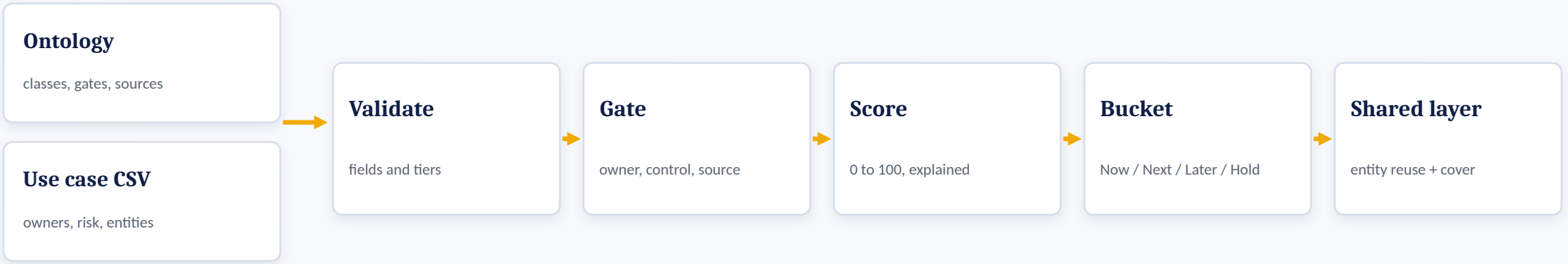
INPUT SCHEMA

One row per use case.

Column	Type	Example
id, name, description	text	UC-001, Policy Q&A Assistant
business_unit, owner	text	Compliance, Aisyah Rahman
data_sources	semicolon list	policy_repository
ai_type	text	RAG
risk_tier	1 to 4	3
value, feasibility, data_readiness	1 to 5	5, 5, 5
effort_weeks	number	6
human_in_loop, monitoring	yes or no	yes, yes
required_entities	semicolon list	policy;regulatory_change

PIPELINE

Validate, gate, score, bucket, shared layer.



Each stage is a pure function. The order matters: gates run before scoring, always.

GATE LOGIC

A gap forces Hold, with the reason attached.

```
for uc in cases:
    gaps = []
    if uc.risk_tier >= 3:
        if not uc.has_named_owner: gaps.append('no named business owner')
        if not uc.human_in_loop:  gaps.append('no human-in-the-loop control')
        if not uc.monitoring:     gaps.append('no monitoring plan')
    unknown = [s for s in uc.data_sources if s not in approved]
    if unknown: gaps.append('unapproved data source: ' + ', '.join(unknown))
    if gaps: bucket = 'Hold'   # a high score cannot rescue this
```

SCORING

Weighted, deterministic, and explainable.

```
score = value/5*40 + readiness/5*25 + feasibility/5*20 + (5-risk)/4*15 - effort/52*10
```

Factor	Contribution	Why it matters
Value	up to 40	The business case.
Data readiness	up to 25	Can we actually run it.
Feasibility	up to 20	Can we build it now.
Risk	up to 15	Lower risk earns more.
Effort	down to -10	Long builds are penalised.

Every score prints its breakdown, so the number is arguable on the evidence, not on the model.

SHARED-LAYER ALGORITHM

Entity reuse, then a greedy cover.

- Build entity_to_cases: for each use case, its required entities.
- Rank entities by reuse (how many use cases need them).
- Greedy cover: repeatedly pick the entity that covers the most uncovered use cases.
- Stop when every use case is covered, or no entity adds coverage.
- Report coverage and any use cases left outside the layer.

Entity	Use cases
customer	7
call_transcript	4
document	4
interaction	4
policy	3

The cover is not the same as the ranking. Customer covers seven cases; policy adds the three that do not touch customer.

OUTPUTS

Three artifacts, one method.

PORTFOLIO REPORT

portfolio-report.md. Summary, ranked table, shared layer, gaps, score breakdowns.

DASHBOARD

portfolio.html. A single file, no dependencies. Four buckets plus the shared-layer panel.

COPILOT SKILL

skills/ai-use-case-portfolio/SKILL.md. The same method, portable to Copilot or any agent.

EXTENSION AND LIMITS

What is easy to add, and what is not proven.

EASY TO EXTEND

- New fields: add to the CSV and ontology together.
- New gates: add to governance_gaps and the ontology.
- New weights: one function, compute_score.
- New sources: a data source to entity mapping.

NOT PROVEN

- The weights are a judgement, not validated.
- Value and readiness are self-reported.
- Risk tiers are simplified.
- It is not a certified AI inventory.

The pattern is the contribution. The numbers need calibration before anyone relies on them.